

Customers

```
use food_delivery_db
switched to db food_delivery_db
db.customers.insertMany([
  {
    customer_id:1,
    name:"Saran",
    email:"saran@gmail.com",
    phone:"9876543210",
    city:"Bangalore"
  },
  {
    customer_id:2,
    name:"Priya",
    email:"priya@gmail.com",
    phone:"9876543211",
    city:"Hyderabad"
  },
  {
    customer_id:3,
    name:"Rahul",
    email:"rahul@gmail.com",
    phone:"9876543212",
    city:"Chennai"
  },
  {
    customer_id:4,
    name:"Deepa",
    email:"deepa@gmail.com",
    phone:"9876543213",
    city:"Pune"
  },
  {
    customer_id:5,
    name:"Anjali",
    email:"anjali@gmail.com",
    phone:"9876543214",
    city:"Bangalore"
  }
]);

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a34164e22e9a5d1382331fd'),
    '1': ObjectId('6a34164e22e9a5d1382331fe'),
    '2': ObjectId('6a34164e22e9a5d1382331ff'),
    '3': ObjectId('6a34164e22e9a5d138233200'),
```

```
      '4': ObjectId('6a34164e22e9a5d138233201')
    }
  }
db.customers.find()
{
  _id: ObjectId('6a34164e22e9a5d1382331fd'),
  customer_id: 1,
  name: 'Saran',
  email: 'saran@gmail.com',
  phone: '9876543210',
  city: 'Bangalore'
}
{
  _id: ObjectId('6a34164e22e9a5d1382331fe'),
  customer_id: 2,
  name: 'Priya',
  email: 'priya@gmail.com',
  phone: '9876543211',
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a34164e22e9a5d1382331ff'),
  customer_id: 3,
  name: 'Rahul',
  email: 'rahul@gmail.com',
  phone: '9876543212',
  city: 'Chennai'
}
{
  _id: ObjectId('6a34164e22e9a5d138233200'),
  customer_id: 4,
  name: 'Deepa',
  email: 'deepa@gmail.com',
  phone: '9876543213',
  city: 'Pune'
}
```

```
}

{
  _id: ObjectId('6a34164e22e9a5d138233201'),
  customer_id: 5,
  name: 'Anjali',
  email: 'anjali@gmail.com',
  phone: '9876543214',
  city: 'Bangalore'
}

db.customers.find({city:"Bangalore"})
{
  _id: ObjectId('6a34164e22e9a5d1382331fd'),
  customer_id: 1,
  name: 'Saran',
  email: 'saran@gmail.com',
  phone: '9876543210',
  city: 'Bangalore'
}

{
  _id: ObjectId('6a34164e22e9a5d138233201'),
  customer_id: 5,
  name: 'Anjali',
  email: 'anjali@gmail.com',
  phone: '9876543214',
  city: 'Bangalore'
}

db.customers.find({city:{$ne:"Hyderabad"}})
{
  _id: ObjectId('6a34164e22e9a5d1382331fd'),
  customer_id: 1,
  name: 'Saran',
  email: 'saran@gmail.com',
  phone: '9876543210',
  city: 'Bangalore'
}
```

```
{
  _id: ObjectId('6a34164e22e9a5d1382331ff'),
  customer_id: 3,
  name: 'Rahul',
  email: 'rahul@gmail.com',
  phone: '9876543212',
  city: 'Chennai'
}
```

```
{
  _id: ObjectId('6a34164e22e9a5d138233200'),
  customer_id: 4,
  name: 'Deepa',
  email: 'deepa@gmail.com',
  phone: '9876543213',
  city: 'Pune'
}
```

```
{
  _id: ObjectId('6a34164e22e9a5d138233201'),
  customer_id: 5,
  name: 'Anjali',
  email: 'anjali@gmail.com',
  phone: '9876543214',
  city: 'Bangalore'
}
```

```
db.customers.find({name:/^S/})
```

```
{
  _id: ObjectId('6a34164e22e9a5d1382331fd'),
  customer_id: 1,
  name: 'Saran',
  email: 'saran@gmail.com',
  phone: '9876543210',
  city: 'Bangalore'
}
```

```
db.customers.find({name:/a$/})
```

```
{
```

```

    _id: ObjectId('6a34164e22e9a5d1382331fe'),
    customer_id: 2,
    name: 'Priya',
    email: 'priya@gmail.com',
    phone: '9876543211',
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a34164e22e9a5d138233200'),
    customer_id: 4,
    name: 'Deepa',
    email: 'deepa@gmail.com',
    phone: '9876543213',
    city: 'Pune'
  }
  db.customers.find({name:/ar/})
  {
    _id: ObjectId('6a34164e22e9a5d1382331fd'),
    customer_id: 1,
    name: 'Saran',
    email: 'saran@gmail.com',
    phone: '9876543210',
    city: 'Bangalore'
  }
  db.customers.find({
    city:{$in:["Bangalore","Chennai"]}
  })
  {
    _id: ObjectId('6a34164e22e9a5d1382331fd'),
    customer_id: 1,
    name: 'Saran',
    email: 'saran@gmail.com',
    phone: '9876543210',
    city: 'Bangalore'
  }
  {

```

```

    _id: ObjectId('6a34164e22e9a5d1382331ff'),
    customer_id: 3,
    name: 'Rahul',
    email: 'rahul@gmail.com',
    phone: '9876543212',
    city: 'Chennai'
  }
  {
    _id: ObjectId('6a34164e22e9a5d138233201'),
    customer_id: 5,
    name: 'Anjali',
    email: 'anjali@gmail.com',
    phone: '9876543214',
    city: 'Bangalore'
  }
  db.customers.find({
    city:"Bangalore",
    name:/^A/
  })
  {
    _id: ObjectId('6a34164e22e9a5d138233201'),
    customer_id: 5,
    name: 'Anjali',
    email: 'anjali@gmail.com',
    phone: '9876543214',
    city: 'Bangalore'
  }
  db.customers.find({
    city:{$in:["Bangalore","Hyderabad","Pune"]}
  })
  {
    _id: ObjectId('6a34164e22e9a5d1382331fd'),
    customer_id: 1,
    name: 'Saran',
    email: 'saran@gmail.com',
    phone: '9876543210',
    city: 'Bangalore'
  }

```

```
}  
  
{  
  _id: ObjectId('6a34164e22e9a5d1382331fe'),  
  customer_id: 2,  
  name: 'Priya',  
  email: 'priya@gmail.com',  
  phone: '9876543211',  
  city: 'Hyderabad'  
}  
  
{  
  _id: ObjectId('6a34164e22e9a5d138233200'),  
  customer_id: 4,  
  name: 'Deepa',  
  email: 'deepa@gmail.com',  
  phone: '9876543213',  
  city: 'Pune'  
}  
  
{  
  _id: ObjectId('6a34164e22e9a5d138233201'),  
  customer_id: 5,  
  name: 'Anjali',  
  email: 'anjali@gmail.com',  
  phone: '9876543214',  
  city: 'Bangalore'  
}  
  
db.customers.find().sort({name:1})  
{  
  _id: ObjectId('6a34164e22e9a5d138233201'),  
  customer_id: 5,  
  name: 'Anjali',  
  email: 'anjali@gmail.com',  
  phone: '9876543214',  
  city: 'Bangalore'  
}  
  
{
```

```
_id: ObjectId('6a34164e22e9a5d138233200'),
customer_id: 4,
name: 'Deepa',
email: 'deepa@gmail.com',
phone: '9876543213',
city: 'Pune'
}
{
  _id: ObjectId('6a34164e22e9a5d1382331fe'),
  customer_id: 2,
  name: 'Priya',
  email: 'priya@gmail.com',
  phone: '9876543211',
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a34164e22e9a5d1382331ff'),
  customer_id: 3,
  name: 'Rahul',
  email: 'rahul@gmail.com',
  phone: '9876543212',
  city: 'Chennai'
}
{
  _id: ObjectId('6a34164e22e9a5d1382331fd'),
  customer_id: 1,
  name: 'Saran',
  email: 'saran@gmail.com',
  phone: '9876543210',
  city: 'Bangalore'
}
db.customers.aggregate([
  {$count:"totalCustomers"}
])
{
```



```

    totalCustomers: 5
  }
  db.customers.aggregate([
    {
      $group:{
        _id:"$city",
        count:{$sum:1}
      }
    }
  ])
  {
    _id: 'Hyderabad',
    count: 1
  }
  {
    _id: 'Bangalore',
    count: 2
  }
  {
    _id: 'Pune',
    count: 1
  }
  {
    _id: 'Chennai',
    count: 1
  }
  db.customers.aggregate([
    {
      $group:{
        _id:"$city",
        count:{$sum:1}
      }
    },
    {$sort:{count:-1}},
    {$limit:1}
  ])
  {
    _id: 'Bangalore',
    count: 2
  }
  db.customers.aggregate([

```

```

    {
      $group:{
        _id:"$city",
        count:{$sum:1}
      }
    },
    {
      $match:{
        count:{$gt:1}
      }
    }
  ])
{
  _id: 'Bangalore',
  count: 2
}

db.customers.aggregate([
  {
    $group:{
      _id:"$city",
      count:{$sum:1}
    }
  },
  {$sort:{count:-1}}
])
{
  _id: 'Bangalore',
  count: 2
}
{
  _id: 'Chennai',
  count: 1
}
{
  _id: 'Hyderabad',
  count: 1
}
{
  _id: 'Pune',
  count: 1
}

```

Restaurants

```
}use food_delivery_db
switched to db food_delivery_db
db.restaurants.insertMany([
{
restaurant_id:101,
restaurant_name:"Pizza Hut",
city:"Bangalore",
rating:4.5
},
{
restaurant_id:102,
restaurant_name:"Dominos",
city:"Hyderabad",
rating:4.3
},
{
restaurant_id:103,
restaurant_name:"Burger King",
city:"Chennai",
rating:4.4
},
{
restaurant_id:104,
restaurant_name:"KFC",
city:"Pune",
rating:4.2
}
]);
```

```
db["Restaurants"].find()
db.restaurants.insertMany([
{
restaurant_id:101,
restaurant_name:"Pizza Hut",
```

```
city:"Bangalore",
rating:4.5
},
{
restaurant_id:102,
restaurant_name:"Dominos",
city:"Hyderabad",
rating:4.3
},
{
restaurant_id:103,
restaurant_name:"Burger King",
city:"Chennai",
rating:4.4
},
{
restaurant_id:104,
restaurant_name:"KFC",
city:"Pune",
rating:4.2
}
]);
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a3416b01254105996ee5a7f'),
    '1': ObjectId('6a3416b01254105996ee5a80'),
    '2': ObjectId('6a3416b01254105996ee5a81'),
    '3': ObjectId('6a3416b01254105996ee5a82')
  }
}

db.restaurants.find()
```

```
{
  _id: ObjectId('6a3416721254105996ee5a7b'),
  restaurant_id: 101,
  restaurant_name: 'Pizza Hut',
  city: 'Bangalore',
  rating: 4.5
}
{
  _id: ObjectId('6a3416721254105996ee5a7c'),
  restaurant_id: 102,
  restaurant_name: 'Dominos',
  city: 'Hyderabad',
  rating: 4.3
}
{
  _id: ObjectId('6a3416721254105996ee5a7d'),
  restaurant_id: 103,
  restaurant_name: 'Burger King',
  city: 'Chennai',
  rating: 4.4
}
{
  _id: ObjectId('6a3416721254105996ee5a7e'),
  restaurant_id: 104,
  restaurant_name: 'KFC',
  city: 'Pune',
  rating: 4.2
}
{
  _id: ObjectId('6a3416b01254105996ee5a7f'),
  restaurant_id: 101,
  restaurant_name: 'Pizza Hut',
  city: 'Bangalore',
  rating: 4.5
}
```

```
}
{
  _id: ObjectId('6a3416b01254105996ee5a80'),
  restaurant_id: 102,
  restaurant_name: 'Dominos',
  city: 'Hyderabad',
  rating: 4.3
}
{
  _id: ObjectId('6a3416b01254105996ee5a81'),
  restaurant_id: 103,
  restaurant_name: 'Burger King',
  city: 'Chennai',
  rating: 4.4
}
{
  _id: ObjectId('6a3416b01254105996ee5a82'),
  restaurant_id: 104,
  restaurant_name: 'KFC',
  city: 'Pune',
  rating: 4.2
}
db.restaurants.find({city:"Bangalore"})
{
  _id: ObjectId('6a3416721254105996ee5a7b'),
  restaurant_id: 101,
  restaurant_name: 'Pizza Hut',
  city: 'Bangalore',
  rating: 4.5
}
{
  _id: ObjectId('6a3416b01254105996ee5a7f'),
  restaurant_id: 101,
  restaurant_name: 'Pizza Hut',
```

```
    city: 'Bangalore',
    rating: 4.5
  }
db.restaurants.find({rating:{$gt:4.3}})
[
  {
    _id: ObjectId('6a3416721254105996ee5a7b'),
    restaurant_id: 101,
    restaurant_name: 'Pizza Hut',
    city: 'Bangalore',
    rating: 4.5
  },
  {
    _id: ObjectId('6a3416721254105996ee5a7d'),
    restaurant_id: 103,
    restaurant_name: 'Burger King',
    city: 'Chennai',
    rating: 4.4
  },
  {
    _id: ObjectId('6a3416b01254105996ee5a7f'),
    restaurant_id: 101,
    restaurant_name: 'Pizza Hut',
    city: 'Bangalore',
    rating: 4.5
  },
  {
    _id: ObjectId('6a3416b01254105996ee5a81'),
    restaurant_id: 103,
    restaurant_name: 'Burger King',
    city: 'Chennai',
    rating: 4.4
  }
]
db.restaurants.find({
  rating:{
```

```
        $gte:4.2,
        $lte:4.5
    }
})
{
  _id: ObjectId('6a3416721254105996ee5a7b'),
  restaurant_id: 101,
  restaurant_name: 'Pizza Hut',
  city: 'Bangalore',
  rating: 4.5
}
{
  _id: ObjectId('6a3416721254105996ee5a7c'),
  restaurant_id: 102,
  restaurant_name: 'Dominos',
  city: 'Hyderabad',
  rating: 4.3
}
{
  _id: ObjectId('6a3416721254105996ee5a7d'),
  restaurant_id: 103,
  restaurant_name: 'Burger King',
  city: 'Chennai',
  rating: 4.4
}
{
  _id: ObjectId('6a3416721254105996ee5a7e'),
  restaurant_id: 104,
  restaurant_name: 'KFC',
  city: 'Pune',
  rating: 4.2
}
{
  _id: ObjectId('6a3416b01254105996ee5a7f'),
```



```
    restaurant_id: 101,  
    restaurant_name: 'Pizza Hut',  
    city: 'Bangalore',  
    rating: 4.5  
}  
{  
  _id: ObjectId('6a3416b01254105996ee5a80'),  
  restaurant_id: 102,  
  restaurant_name: 'Dominos',  
  city: 'Hyderabad',  
  rating: 4.3  
}  
{  
  _id: ObjectId('6a3416b01254105996ee5a81'),  
  restaurant_id: 103,  
  restaurant_name: 'Burger King',  
  city: 'Chennai',  
  rating: 4.4  
}  
{  
  _id: ObjectId('6a3416b01254105996ee5a82'),  
  restaurant_id: 104,  
  restaurant_name: 'KFC',  
  city: 'Pune',  
  rating: 4.2  
}  
db.restaurants.find({  
  restaurant_name:/^P/  
})  
{  
  _id: ObjectId('6a3416721254105996ee5a7b'),  
  restaurant_id: 101,  
  restaurant_name: 'Pizza Hut',  
  city: 'Bangalore',
```

```
    rating: 4.5
  }
  {
    _id: ObjectId('6a3416b01254105996ee5a7f'),
    restaurant_id: 101,
    restaurant_name: 'Pizza Hut',
    city: 'Bangalore',
    rating: 4.5
  }
  db.restaurants.find({
    restaurant_name:/King/
  })
  {
    _id: ObjectId('6a3416721254105996ee5a7d'),
    restaurant_id: 103,
    restaurant_name: 'Burger King',
    city: 'Chennai',
    rating: 4.4
  }
  {
    _id: ObjectId('6a3416b01254105996ee5a81'),
    restaurant_id: 103,
    restaurant_name: 'Burger King',
    city: 'Chennai',
    rating: 4.4
  }
  db.restaurants.find({
    city:{$in:["Chennai","Hyderabad"]}
  })
  {
    _id: ObjectId('6a3416721254105996ee5a7c'),
    restaurant_id: 102,
    restaurant_name: 'Dominos',
    city: 'Hyderabad',
```

```
    rating: 4.3
  }
  {
    _id: ObjectId('6a3416721254105996ee5a7d'),
    restaurant_id: 103,
    restaurant_name: 'Burger King',
    city: 'Chennai',
    rating: 4.4
  }
  {
    _id: ObjectId('6a3416b01254105996ee5a80'),
    restaurant_id: 102,
    restaurant_name: 'Dominos',
    city: 'Hyderabad',
    rating: 4.3
  }
  {
    _id: ObjectId('6a3416b01254105996ee5a81'),
    restaurant_id: 103,
    restaurant_name: 'Burger King',
    city: 'Chennai',
    rating: 4.4
  }
  db.restaurants.find({
    city:{$ne:"Pune"}
  })
  {
    _id: ObjectId('6a3416721254105996ee5a7b'),
    restaurant_id: 101,
    restaurant_name: 'Pizza Hut',
    city: 'Bangalore',
    rating: 4.5
  }
  {
```

```
_id: ObjectId('6a3416721254105996ee5a7c'),
restaurant_id: 102,
restaurant_name: 'Dominos',
city: 'Hyderabad',
rating: 4.3
}
{
_id: ObjectId('6a3416721254105996ee5a7d'),
restaurant_id: 103,
restaurant_name: 'Burger King',
city: 'Chennai',
rating: 4.4
}
{
_id: ObjectId('6a3416b01254105996ee5a7f'),
restaurant_id: 101,
restaurant_name: 'Pizza Hut',
city: 'Bangalore',
rating: 4.5
}
{
_id: ObjectId('6a3416b01254105996ee5a80'),
restaurant_id: 102,
restaurant_name: 'Dominos',
city: 'Hyderabad',
rating: 4.3
}
{
_id: ObjectId('6a3416b01254105996ee5a81'),
restaurant_id: 103,
restaurant_name: 'Burger King',
city: 'Chennai',
rating: 4.4
}
```

```
db.restaurants.find().sort({rating:-1})

{
  _id: ObjectId('6a3416721254105996ee5a7b'),
  restaurant_id: 101,
  restaurant_name: 'Pizza Hut',
  city: 'Bangalore',
  rating: 4.5
}

{
  _id: ObjectId('6a3416b01254105996ee5a7f'),
  restaurant_id: 101,
  restaurant_name: 'Pizza Hut',
  city: 'Bangalore',
  rating: 4.5
}

{
  _id: ObjectId('6a3416721254105996ee5a7d'),
  restaurant_id: 103,
  restaurant_name: 'Burger King',
  city: 'Chennai',
  rating: 4.4
}

{
  _id: ObjectId('6a3416b01254105996ee5a81'),
  restaurant_id: 103,
  restaurant_name: 'Burger King',
  city: 'Chennai',
  rating: 4.4
}

{
  _id: ObjectId('6a3416721254105996ee5a7c'),
  restaurant_id: 102,
  restaurant_name: 'Dominos',
  city: 'Hyderabad',
```

```
    rating: 4.3
  }
  {
    _id: ObjectId('6a3416b01254105996ee5a80'),
    restaurant_id: 102,
    restaurant_name: 'Dominos',
    city: 'Hyderabad',
    rating: 4.3
  }
  {
    _id: ObjectId('6a3416721254105996ee5a7e'),
    restaurant_id: 104,
    restaurant_name: 'KFC',
    city: 'Pune',
    rating: 4.2
  }
  {
    _id: ObjectId('6a3416b01254105996ee5a82'),
    restaurant_id: 104,
    restaurant_name: 'KFC',
    city: 'Pune',
    rating: 4.2
  }
db.restaurants.find().sort({rating:-1}).limit(3)
{
  _id: ObjectId('6a3416b01254105996ee5a7f'),
  restaurant_id: 101,
  restaurant_name: 'Pizza Hut',
  city: 'Bangalore',
  rating: 4.5
}
{
  _id: ObjectId('6a3416721254105996ee5a7b'),
  restaurant_id: 101,
```

```

    restaurant_name: 'Pizza Hut',
    city: 'Bangalore',
    rating: 4.5
  }
  {
    _id: ObjectId('6a3416721254105996ee5a7d'),
    restaurant_id: 103,
    restaurant_name: 'Burger King',
    city: 'Chennai',
    rating: 4.4
  }
db.restaurants.aggregate([
  {
    $group:{
      _id:null,
      avgRating:{$avg:"$rating"}
    }
  }
])
{
  _id: null,
  avgRating: 4.35
}
db.restaurants.aggregate([
  {
    $group:{
      _id:null,
      maxRating:{$max:"$rating"}
    }
  }
])
{
  _id: null,
  maxRating: 4.5
}

```

```

}
db.restaurants.aggregate([
  {
    $group:{
      _id:null,
      minRating:{$min:"$rating"}
    }
  }
])

```

```

{
  _id: null,
  minRating: 4.2
}

```

```

db.restaurants.aggregate([
  {
    $group:{
      _id:"$city",
      count:{$sum:1}
    }
  }
])

```

```

{
  _id: 'Bangalore',
  count: 2
}

```

```

{
  _id: 'Chennai',
  count: 2
}

```

```

{
  _id: 'Pune',
  count: 2
}

```

```

{

```



```
  _id: 'Hyderabad',
  count: 2
}
db.restaurants.aggregate([
  {
    $group:{
      _id:"$city",
      count:{$sum:1}
    }
  },
  {
    $match:{
      count:{$gt:1}
    }
  }
])
{
  _id: 'Hyderabad',
  count: 2
}
{
  _id: 'Bangalore',
  count: 2
}
{
  _id: 'Pune',
  count: 2
}
{
  _id: 'Chennai',
  count: 2
}
```

Orders

```
use food_delivery_db

switched to db food_delivery_db

db.orders.insertMany([

{
  order_id:1001,
  customer_id:1,
  restaurant_id:101,
  items:[
    {
      item_name:"Veg Pizza",
      quantity:2,
      price:250
    }
  ],
  total_amount:500,
  status:"Delivered"
},
{
  order_id:1002,
  customer_id:2,
  restaurant_id:102,
  items:[
    {
      item_name:"Cheese Burst Pizza",
      quantity:1,
      price:300
    }
  ],
  total_amount:300,
  status:"Pending"
},
{
```

```
order_id:1003,
customer_id:3,
restaurant_id:103,
items:[
{
item_name:"Veg Burger",
quantity:2,
price:180
}
],
total_amount:360,
status:"Delivered"
}
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a34170588248e90566a7677'),
    '1': ObjectId('6a34170588248e90566a7678'),
    '2': ObjectId('6a34170588248e90566a7679')
  }
}
db.orders.find()
{
  _id: ObjectId('6a34170588248e90566a7677'),
  order_id: 1001,
  customer_id: 1,
  restaurant_id: 101,
  items: [
    {
      item_name: 'Veg Pizza',
      quantity: 2,
      price: 250
    }
  ]
}
```

```
],
total_amount: 500,
status: 'Delivered'
}
{
  _id: ObjectId('6a34170588248e90566a7678'),
  order_id: 1002,
  customer_id: 2,
  restaurant_id: 102,
  items: [
    {
      item_name: 'Cheese Burst Pizza',
      quantity: 1,
      price: 300
    }
  ],
total_amount: 300,
status: 'Pending'
}
{
  _id: ObjectId('6a34170588248e90566a7679'),
  order_id: 1003,
  customer_id: 3,
  restaurant_id: 103,
  items: [
    {
      item_name: 'Veg Burger',
      quantity: 2,
      price: 180
    }
  ],
total_amount: 360,
status: 'Delivered'
}
```

```
db.orders.find({
  status:"Delivered"
})
{
  _id: ObjectId('6a34170588248e90566a7677'),
  order_id: 1001,
  customer_id: 1,
  restaurant_id: 101,
  items: [
    {
      item_name: 'Veg Pizza',
      quantity: 2,
      price: 250
    }
  ],
  total_amount: 500,
  status: 'Delivered'
}
{
  _id: ObjectId('6a34170588248e90566a7679'),
  order_id: 1003,
  customer_id: 3,
  restaurant_id: 103,
  items: [
    {
      item_name: 'Veg Burger',
      quantity: 2,
      price: 180
    }
  ],
  total_amount: 360,
  status: 'Delivered'
}
db.orders.find({
```

```
        status:"Pending"
    })
    {
      _id: ObjectId('6a34170588248e90566a7678'),
      order_id: 1002,
      customer_id: 2,
      restaurant_id: 102,
      items: [
        {
          item_name: 'Cheese Burst Pizza',
          quantity: 1,
          price: 300
        }
      ],
      total_amount: 300,
      status: 'Pending'
    }
    db.orders.find({
      total_amount:{$gt:400}
    })
    {
      _id: ObjectId('6a34170588248e90566a7677'),
      order_id: 1001,
      customer_id: 1,
      restaurant_id: 101,
      items: [
        {
          item_name: 'Veg Pizza',
          quantity: 2,
          price: 250
        }
      ],
      total_amount: 500,
      status: 'Delivered'
    }
  ]
}
```

```
}
db.orders.find({
  total_amount:{
    $gte:300,
    $lte:600
  }
})
{
  _id: ObjectId('6a34170588248e90566a7677'),
  order_id: 1001,
  customer_id: 1,
  restaurant_id: 101,
  items: [
    {
      item_name: 'Veg Pizza',
      quantity: 2,
      price: 250
    }
  ],
  total_amount: 500,
  status: 'Delivered'
}
{
  _id: ObjectId('6a34170588248e90566a7678'),
  order_id: 1002,
  customer_id: 2,
  restaurant_id: 102,
  items: [
    {
      item_name: 'Cheese Burst Pizza',
      quantity: 1,
      price: 300
    }
  ],
}
```

```
total_amount: 300,
status: 'Pending'
}
{
  _id: ObjectId('6a34170588248e90566a7679'),
  order_id: 1003,
  customer_id: 3,
  restaurant_id: 103,
  items: [
    {
      item_name: 'Veg Burger',
      quantity: 2,
      price: 180
    }
  ],
  total_amount: 360,
  status: 'Delivered'
}
db.orders.find({
  customer_id:1
})
{
  _id: ObjectId('6a34170588248e90566a7677'),
  order_id: 1001,
  customer_id: 1,
  restaurant_id: 101,
  items: [
    {
      item_name: 'Veg Pizza',
      quantity: 2,
      price: 250
    }
  ],
  total_amount: 500,
```



```
    status: 'Delivered'
  }
  db.orders.find().sort({
    total_amount:-1
  })
  {
    _id: ObjectId('6a34170588248e90566a7677'),
    order_id: 1001,
    customer_id: 1,
    restaurant_id: 101,
    items: [
      {
        item_name: 'Veg Pizza',
        quantity: 2,
        price: 250
      }
    ],
    total_amount: 500,
    status: 'Delivered'
  }
  {
    _id: ObjectId('6a34170588248e90566a7679'),
    order_id: 1003,
    customer_id: 3,
    restaurant_id: 103,
    items: [
      {
        item_name: 'Veg Burger',
        quantity: 2,
        price: 180
      }
    ],
    total_amount: 360,
    status: 'Delivered'
```

```

}
{
  _id: ObjectId('6a34170588248e90566a7678'),
  order_id: 1002,
  customer_id: 2,
  restaurant_id: 102,
  items: [
    {
      item_name: 'Cheese Burst Pizza',
      quantity: 1,
      price: 300
    }
  ],
  total_amount: 300,
  status: 'Pending'
}
db.orders.find()
.sort({total_amount:-1})
.limit(3)
{
  _id: ObjectId('6a34170588248e90566a7677'),
  order_id: 1001,
  customer_id: 1,
  restaurant_id: 101,
  items: [
    {
      item_name: 'Veg Pizza',
      quantity: 2,
      price: 250
    }
  ],
  total_amount: 500,
  status: 'Delivered'
}

```

```

{
  _id: ObjectId('6a34170588248e90566a7679'),
  order_id: 1003,
  customer_id: 3,
  restaurant_id: 103,
  items: [
    {
      item_name: 'Veg Burger',
      quantity: 2,
      price: 180
    }
  ],
  total_amount: 360,
  status: 'Delivered'
}
{
  _id: ObjectId('6a34170588248e90566a7678'),
  order_id: 1002,
  customer_id: 2,
  restaurant_id: 102,
  items: [
    {
      item_name: 'Cheese Burst Pizza',
      quantity: 1,
      price: 300
    }
  ],
  total_amount: 300,
  status: 'Pending'
}
db.orders.find({
  status: {$ne:"Delivered"}
})
{

```

```
_id: ObjectId('6a34170588248e90566a7678'),
order_id: 1002,
customer_id: 2,
restaurant_id: 102,
items: [
  {
    item_name: 'Cheese Burst Pizza',
    quantity: 1,
    price: 300
  }
],
total_amount: 300,
status: 'Pending'
}
db.orders.find({
  total_amount:{$gt:300},
  status:"Delivered"
})
{
  _id: ObjectId('6a34170588248e90566a7677'),
  order_id: 1001,
  customer_id: 1,
  restaurant_id: 101,
  items: [
    {
      item_name: 'Veg Pizza',
      quantity: 2,
      price: 250
    }
  ],
  total_amount: 500,
  status: 'Delivered'
}
{
```

```

    _id: ObjectId('6a34170588248e90566a7679'),
    order_id: 1003,
    customer_id: 3,
    restaurant_id: 103,
    items: [
      {
        item_name: 'Veg Burger',
        quantity: 2,
        price: 180
      }
    ],
    total_amount: 360,
    status: 'Delivered'
  }
}

db.orders.aggregate([
  {
    $group:{
      _id:null,
      totalRevenue:{$sum:"$total_amount"}
    }
  }
])

{
  _id: null,
  totalRevenue: 1160
}

db.orders.aggregate([
  {
    $group:{
      _id:null,
      avgAmount:{$avg:"$total_amount"}
    }
  }
])

```

```

{
  _id: null,
  avgAmount: 386.6666666666667
}
db.orders.aggregate([
  {
    $group:{
      _id:null,
      maxAmount:{$max:"$total_amount"}
    }
  }
])
{
  _id: null,
  maxAmount: 500
}
db.orders.aggregate([
  {
    $group:{
      _id:null,
      minAmount:{$min:"$total_amount"}
    }
  }
])
{
  _id: null,
  minAmount: 300
}
db.orders.aggregate([
  {
    $group:{
      _id:"$status",
      count:{$sum:1}
    }
  }
])

```

```

    }
  ])
  {
    _id: 'Delivered',
    count: 2
  }
  {
    _id: 'Pending',
    count: 1
  }
  db.orders.aggregate([
    {
      $group:{
        _id:"$status",
        count:{$sum:1}
      }
    },
    {
      $match:{
        count:{$gt:1}
      }
    }
  ])
  {
    _id: 'Delivered',
    count: 2
  }
  db.orders.aggregate([
    {
      $group:{
        _id:"$status",
        count:{$sum:1}
      }
    },

```

```

    {
      $sort:{
        count:-1
      }
    }
  ])
  {
    _id: 'Delivered',
    count: 2
  }
  {
    _id: 'Pending',
    count: 1
  }
db.orders.aggregate([
  {
    $match:{
      status:"Delivered"
    }
  },
  {
    $group:{
      _id:null,
      totalRevenue:{$sum:"$total_amount"}
    }
  }
])
{
  _id: null,
  totalRevenue: 860
}
db.orders.aggregate([
  {
    $match:{

```



```

        status:"Delivered"
    }
},
{
    $group:{
        _id:null,
        avgAmount:{$avg:"$total_amount"}
    }
}
])
{
    _id: null,
    avgAmount: 430
}
db.orders.aggregate([
    {
        $group:{
            _id:"$customer_id",
            orderCount:{$sum:1}
        }
    }
])
{
    _id: 1,
    orderCount: 1
}
{
    _id: 2,
    orderCount: 1
}
{
    _id: 3,
    orderCount: 1
}
}

```

Menu

```
use food_delivery_db

switched to db food_delivery_db

db.menu.insertMany([

{
item_id:1,
restaurant_id:101,
item_name:"Veg Pizza",
price:250
},

{
item_id:2,
restaurant_id:101,
item_name:"Chicken Pizza",
price:350
},

{
item_id:3,
restaurant_id:102,
item_name:"Cheese Burst Pizza",
price:300
},

{
item_id:4,
restaurant_id:103,
item_name:"Veg Burger",
price:180
},

{
item_id:5,
restaurant_id:104,
item_name:"Chicken Bucket",
price:550
}
```

```
}
]);

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a34169da9f06094916014df'),
    '1': ObjectId('6a34169da9f06094916014e0'),
    '2': ObjectId('6a34169da9f06094916014e1'),
    '3': ObjectId('6a34169da9f06094916014e2'),
    '4': ObjectId('6a34169da9f06094916014e3')
  }
}

db.menu.find()

{
  _id: ObjectId('6a34169da9f06094916014df'),
  item_id: 1,
  restaurant_id: 101,
  item_name: 'Veg Pizza',
  price: 250
}

{
  _id: ObjectId('6a34169da9f06094916014e0'),
  item_id: 2,
  restaurant_id: 101,
  item_name: 'Chicken Pizza',
  price: 350
}

{
  _id: ObjectId('6a34169da9f06094916014e1'),
  item_id: 3,
  restaurant_id: 102,
  item_name: 'Cheese Burst Pizza',
  price: 300
}
```

```
}
{
  _id: ObjectId('6a34169da9f06094916014e2'),
  item_id: 4,
  restaurant_id: 103,
  item_name: 'Veg Burger',
  price: 180
}
{
  _id: ObjectId('6a34169da9f06094916014e3'),
  item_id: 5,
  restaurant_id: 104,
  item_name: 'Chicken Bucket',
  price: 550
}
db.menu.find({
  price:{$gt:250}
})
{
  _id: ObjectId('6a34169da9f06094916014e0'),
  item_id: 2,
  restaurant_id: 101,
  item_name: 'Chicken Pizza',
  price: 350
}
{
  _id: ObjectId('6a34169da9f06094916014e1'),
  item_id: 3,
  restaurant_id: 102,
  item_name: 'Cheese Burst Pizza',
  price: 300
}
{
  _id: ObjectId('6a34169da9f06094916014e3'),
```

```
    item_id: 5,
    restaurant_id: 104,
    item_name: 'Chicken Bucket',
    price: 550
  }
  db.menu.find({
    price:{
      $gte:200,
      $lte:400
    }
  })
  {
    _id: ObjectId('6a34169da9f06094916014df'),
    item_id: 1,
    restaurant_id: 101,
    item_name: 'Veg Pizza',
    price: 250
  }
  {
    _id: ObjectId('6a34169da9f06094916014e0'),
    item_id: 2,
    restaurant_id: 101,
    item_name: 'Chicken Pizza',
    price: 350
  }
  {
    _id: ObjectId('6a34169da9f06094916014e1'),
    item_id: 3,
    restaurant_id: 102,
    item_name: 'Cheese Burst Pizza',
    price: 300
  }
  db.menu.find({
    restaurant_id:101
```

```
  })
  {
    _id: ObjectId('6a34169da9f06094916014df'),
    item_id: 1,
    restaurant_id: 101,
    item_name: 'Veg Pizza',
    price: 250
  }
  {
    _id: ObjectId('6a34169da9f06094916014e0'),
    item_id: 2,
    restaurant_id: 101,
    item_name: 'Chicken Pizza',
    price: 350
  }
  db.menu.find({
    item_name: /^C/
  })
  {
    _id: ObjectId('6a34169da9f06094916014e0'),
    item_id: 2,
    restaurant_id: 101,
    item_name: 'Chicken Pizza',
    price: 350
  }
  {
    _id: ObjectId('6a34169da9f06094916014e1'),
    item_id: 3,
    restaurant_id: 102,
    item_name: 'Cheese Burst Pizza',
    price: 300
  }
  {
    _id: ObjectId('6a34169da9f06094916014e3'),
```

```
    item_id: 5,
    restaurant_id: 104,
    item_name: 'Chicken Bucket',
    price: 550
  }
  db.menu.find({
    item_name:/Pizza/
  })
  {
    _id: ObjectId('6a34169da9f06094916014df'),
    item_id: 1,
    restaurant_id: 101,
    item_name: 'Veg Pizza',
    price: 250
  }
  {
    _id: ObjectId('6a34169da9f06094916014e0'),
    item_id: 2,
    restaurant_id: 101,
    item_name: 'Chicken Pizza',
    price: 350
  }
  {
    _id: ObjectId('6a34169da9f06094916014e1'),
    item_id: 3,
    restaurant_id: 102,
    item_name: 'Cheese Burst Pizza',
    price: 300
  }
  db.menu.find().sort({price:1})
  {
    _id: ObjectId('6a34169da9f06094916014e2'),
    item_id: 4,
    restaurant_id: 103,
```

```
    item_name: 'Veg Burger',
    price: 180
  }
  {
    _id: ObjectId('6a34169da9f06094916014df'),
    item_id: 1,
    restaurant_id: 101,
    item_name: 'Veg Pizza',
    price: 250
  }
  {
    _id: ObjectId('6a34169da9f06094916014e1'),
    item_id: 3,
    restaurant_id: 102,
    item_name: 'Cheese Burst Pizza',
    price: 300
  }
  {
    _id: ObjectId('6a34169da9f06094916014e0'),
    item_id: 2,
    restaurant_id: 101,
    item_name: 'Chicken Pizza',
    price: 350
  }
  {
    _id: ObjectId('6a34169da9f06094916014e3'),
    item_id: 5,
    restaurant_id: 104,
    item_name: 'Chicken Bucket',
    price: 550
  }
  db.menu.find().sort({price:-1}).limit(3)
  {
    _id: ObjectId('6a34169da9f06094916014e3'),
```



```
    item_id: 5,
    restaurant_id: 104,
    item_name: 'Chicken Bucket',
    price: 550
  }
  {
    _id: ObjectId('6a34169da9f06094916014e0'),
    item_id: 2,
    restaurant_id: 101,
    item_name: 'Chicken Pizza',
    price: 350
  }
  {
    _id: ObjectId('6a34169da9f06094916014e1'),
    item_id: 3,
    restaurant_id: 102,
    item_name: 'Cheese Burst Pizza',
    price: 300
  }
  db.menu.find({
    price:{$lt:300}
  })
  {
    _id: ObjectId('6a34169da9f06094916014df'),
    item_id: 1,
    restaurant_id: 101,
    item_name: 'Veg Pizza',
    price: 250
  }
  {
    _id: ObjectId('6a34169da9f06094916014e2'),
    item_id: 4,
    restaurant_id: 103,
    item_name: 'Veg Burger',
```

```
    price: 180
  }
  db.menu.find({
    price:{$ne:350}
  })
  {
    _id: ObjectId('6a34169da9f06094916014df'),
    item_id: 1,
    restaurant_id: 101,
    item_name: 'Veg Pizza',
    price: 250
  }
  {
    _id: ObjectId('6a34169da9f06094916014e1'),
    item_id: 3,
    restaurant_id: 102,
    item_name: 'Cheese Burst Pizza',
    price: 300
  }
  {
    _id: ObjectId('6a34169da9f06094916014e2'),
    item_id: 4,
    restaurant_id: 103,
    item_name: 'Veg Burger',
    price: 180
  }
  {
    _id: ObjectId('6a34169da9f06094916014e3'),
    item_id: 5,
    restaurant_id: 104,
    item_name: 'Chicken Bucket',
    price: 550
  }
  db.menu.aggregate([
```

```

    {
      $group:{
        _id:null,
        avgPrice:{$avg:"$price"}
      }
    }
  ])
  {
    _id: null,
    avgPrice: 326
  }
  db.menu.aggregate([
    {
      $group:{
        _id:null,
        maxPrice:{$max:"$price"}
      }
    }
  ])
  {
    _id: null,
    maxPrice: 550
  }
  db.menu.aggregate([
    {
      $group:{
        _id:null,
        minPrice:{$min:"$price"}
      }
    }
  ])
  {
    _id: null,
    minPrice: 180
  }

```

```

}
db.menu.aggregate([
  {
    $group:{
      _id:null,
      totalPrice:{$sum:"$price"}
    }
  }
])

```

```

{
  _id: null,
  totalPrice: 1630
}

```

```

db.menu.aggregate([
  {
    $group:{
      _id:"$restaurant_id",
      itemCount:{$sum:1}
    }
  }
])

```

```

{
  _id: 103,
  itemCount: 1
}

```

```

{
  _id: 102,
  itemCount: 1
}

```

```

{
  _id: 104,
  itemCount: 1
}

```

```

{

```

```
_id: 101,  
itemCount: 2  
}
```

Payments

```
use food_delivery_db  
switched to db food_delivery_db  
db.payments.insertMany([  
  {  
    payment_id:501,  
    order_id:1001,  
    payment_method:"UPI",  
    amount:500,  
    status:"Success"  
  },  
  {  
    payment_id:502,  
    order_id:1002,  
    payment_method:"Credit Card",  
    amount:300,  
    status:"Pending"  
  },  
  {  
    payment_id:503,  
    order_id:1003,  
    payment_method:"Cash",  
    amount:360,  
    status:"Success"  
  }  
]);  
  
{  
  acknowledged: true,  
  insertedIds: {
```

```

    '0': ObjectId('6a3417fd1c91f3c5612ca9b3'),
    '1': ObjectId('6a3417fd1c91f3c5612ca9b4'),
    '2': ObjectId('6a3417fd1c91f3c5612ca9b5')
  }
}
db.payments.find()
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b3'),
  payment_id: 501,
  order_id: 1001,
  payment_method: 'UPI',
  amount: 500,
  status: 'Success'
}
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b4'),
  payment_id: 502,
  order_id: 1002,
  payment_method: 'Credit Card',
  amount: 300,
  status: 'Pending'
}
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b5'),
  payment_id: 503,
  order_id: 1003,
  payment_method: 'Cash',
  amount: 360,
  status: 'Success'
}
db.payments.find({
  status:"Success"
})
{

```

```
_id: ObjectId('6a3417fd1c91f3c5612ca9b3'),
payment_id: 501,
order_id: 1001,
payment_method: 'UPI',
amount: 500,
status: 'Success'
}
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b5'),
  payment_id: 503,
  order_id: 1003,
  payment_method: 'Cash',
  amount: 360,
  status: 'Success'
}
db.payments.find({
  status:"Pending"
})
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b4'),
  payment_id: 502,
  order_id: 1002,
  payment_method: 'Credit Card',
  amount: 300,
  status: 'Pending'
}
db.payments.find({
  payment_method:"UPI"
})
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b3'),
  payment_id: 501,
  order_id: 1001,
  payment_method: 'UPI',
```

```
    amount: 500,
    status: 'Success'
  }
  db.payments.find({
    amount:{$gt:300}
  })
  {
    _id: ObjectId('6a3417fd1c91f3c5612ca9b3'),
    payment_id: 501,
    order_id: 1001,
    payment_method: 'UPI',
    amount: 500,
    status: 'Success'
  }
  {
    _id: ObjectId('6a3417fd1c91f3c5612ca9b5'),
    payment_id: 503,
    order_id: 1003,
    payment_method: 'Cash',
    amount: 360,
    status: 'Success'
  }
  db.payments.find().sort({
    amount:1
  })
  {
    _id: ObjectId('6a3417fd1c91f3c5612ca9b4'),
    payment_id: 502,
    order_id: 1002,
    payment_method: 'Credit Card',
    amount: 300,
    status: 'Pending'
  }
  {
```



```
_id: ObjectId('6a3417fd1c91f3c5612ca9b5'),
payment_id: 503,
order_id: 1003,
payment_method: 'Cash',
amount: 360,
status: 'Success'
}
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b3'),
  payment_id: 501,
  order_id: 1001,
  payment_method: 'UPI',
  amount: 500,
  status: 'Success'
}
db.payments.find()
.sort({amount:-1})
.limit(3)
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b3'),
  payment_id: 501,
  order_id: 1001,
  payment_method: 'UPI',
  amount: 500,
  status: 'Success'
}
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b5'),
  payment_id: 503,
  order_id: 1003,
  payment_method: 'Cash',
  amount: 360,
  status: 'Success'
}
```

```
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b4'),
  payment_id: 502,
  order_id: 1002,
  payment_method: 'Credit Card',
  amount: 300,
  status: 'Pending'
}
db.payments.find({
  status:{$ne:"Success"}
})
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b4'),
  payment_id: 502,
  order_id: 1002,
  payment_method: 'Credit Card',
  amount: 300,
  status: 'Pending'
}
db.payments.find({
  payment_method:{
    $in:["Cash","UPI"]
  }
})
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b3'),
  payment_id: 501,
  order_id: 1001,
  payment_method: 'UPI',
  amount: 500,
  status: 'Success'
}
{
  _id: ObjectId('6a3417fd1c91f3c5612ca9b5'),
```

```
    payment_id: 503,  
    order_id: 1003,  
    payment_method: 'Cash',  
    amount: 360,  
    status: 'Success'  
}  
db.payments.find({  
  amount:{  
    $gte:300,  
    $lte:500  
  }  
})  
{  
  _id: ObjectId('6a3417fd1c91f3c5612ca9b3'),  
  payment_id: 501,  
  order_id: 1001,  
  payment_method: 'UPI',  
  amount: 500,  
  status: 'Success'  
}  
{  
  _id: ObjectId('6a3417fd1c91f3c5612ca9b4'),  
  payment_id: 502,  
  order_id: 1002,  
  payment_method: 'Credit Card',  
  amount: 300,  
  status: 'Pending'  
}  
{  
  _id: ObjectId('6a3417fd1c91f3c5612ca9b5'),  
  payment_id: 503,  
  order_id: 1003,  
  payment_method: 'Cash',  
  amount: 360,
```

```

    status: 'Success'
  }
  db.payments.aggregate([
    {
      $group:{
        _id:null,
        totalAmount:{$sum:"$amount"}
      }
    }
  ])
  {
    _id: null,
    totalAmount: 1160
  }
  db.payments.aggregate([
    {
      $group:{
        _id:null,
        avgAmount:{$avg:"$amount"}
      }
    }
  ])
  {
    _id: null,
    avgAmount: 386.66666666666667
  }
  db.payments.aggregate([
    {
      $group:{
        _id:null,
        maxAmount:{$max:"$amount"}
      }
    }
  ])

```

```

{
  _id: null,
  maxAmount: 500
}
db.payments.aggregate([
  {
    $group:{
      _id:null,
      minAmount:{$min:"$amount"}
    }
  }
])
{
  _id: null,
  minAmount: 300
}
db.payments.aggregate([
  {
    $group:{
      _id:"$payment_method",
      count:{$sum:1}
    }
  }
])
{
  _id: 'Cash',
  count: 1
}
{
  _id: 'Credit Card',
  count: 1
}
{
  _id: 'UPI',

```

```

    count: 1
  }
  db.payments.aggregate([
    {
      $group:{
        _id:"$status",
        count:{$sum:1}
      }
    }
  ])
  {
    _id: 'Pending',
    count: 1
  }
  {
    _id: 'Success',
    count: 2
  }
  db.payments.aggregate([
    {
      $group:{
        _id:"$payment_method",
        count:{$sum:1}
      }
    },
    {
      $match:{
        count:{$gt:1}
      }
    }
  ])
  db.payments.aggregate([
    {
      $group:{

```

```

        _id:"$payment_method",
        count:{$sum:1}
    }
},
{
    $match:{
        count:{$gt:1}
    }
}
])
db.payments.aggregate([
{
    $match:{
        payment_method:"UPI"
    }
},
{
    $group:{
        _id:null,
        totalAmount:{$sum:"$amount"}
    }
}
])
{
    _id: null,
    totalAmount: 500
}
db.payments.aggregate([
{
    $match:{
        payment_method:"Credit Card"
    }
},
{

```

```

    $group:{
      _id:null,
      avgAmount:{$avg:"$amount"}
    }
  }
])
{
  _id: null,
  avgAmount: 300
}
db.payments.aggregate([
  {
    $group:{
      _id:"$payment_method",
      totalAmount:{$sum:"$amount"}
    }
  },
  {
    $sort:{
      totalAmount:-1
    }
  }
])
{
  _id: 'UPI',
  totalAmount: 500
}
{
  _id: 'Cash',
  totalAmount: 360
}
{
  _id: 'Credit Card',
  totalAmount: 300
}

```



```
}
```

Delivery Agents

```
use food_delivery_db
switched to db food_delivery_db
db.delivery_agents.insertMany([
  {
    agent_id:1,
    agent_name:"Kumar",
    city:"Bangalore",
    rating:4.8
  },
  {
    agent_id:2,
    agent_name:"Arjun",
    city:"Hyderabad",
    rating:4.6
  },
  {
    agent_id:3,
    agent_name:"Sneha",
    city:"Chennai",
    rating:4.9
  }
]);

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a34182b76d019eb215bcd44'),
    '1': ObjectId('6a34182b76d019eb215bcd45'),
    '2': ObjectId('6a34182b76d019eb215bcd46')
  }
}
```

```
db.delivery_agents.find()

{
  _id: ObjectId('6a34182b76d019eb215bcd44'),
  agent_id: 1,
  agent_name: 'Kumar',
  city: 'Bangalore',
  rating: 4.8
}

{
  _id: ObjectId('6a34182b76d019eb215bcd45'),
  agent_id: 2,
  agent_name: 'Arjun',
  city: 'Hyderabad',
  rating: 4.6
}

{
  _id: ObjectId('6a34182b76d019eb215bcd46'),
  agent_id: 3,
  agent_name: 'Sneha',
  city: 'Chennai',
  rating: 4.9
}

db.delivery_agents.find({
  city:"Bangalore"
})

{
  _id: ObjectId('6a34182b76d019eb215bcd44'),
  agent_id: 1,
  agent_name: 'Kumar',
  city: 'Bangalore',
  rating: 4.8
}

db.delivery_agents.find({
  rating:{>4.7}
```

```
  })
  {
    _id: ObjectId('6a34182b76d019eb215bcd44'),
    agent_id: 1,
    agent_name: 'Kumar',
    city: 'Bangalore',
    rating: 4.8
  }
  {
    _id: ObjectId('6a34182b76d019eb215bcd46'),
    agent_id: 3,
    agent_name: 'Sneha',
    city: 'Chennai',
    rating: 4.9
  }
  db.delivery_agents.find({
    agent_name:/^A/
  })
  {
    _id: ObjectId('6a34182b76d019eb215bcd45'),
    agent_id: 2,
    agent_name: 'Arjun',
    city: 'Hyderabad',
    rating: 4.6
  }
  db.delivery_agents.find().sort({
    rating:1
  })
  {
    _id: ObjectId('6a34182b76d019eb215bcd45'),
    agent_id: 2,
    agent_name: 'Arjun',
    city: 'Hyderabad',
    rating: 4.6
  }
```

```

}
{
  _id: ObjectId('6a34182b76d019eb215bcd44'),
  agent_id: 1,
  agent_name: 'Kumar',
  city: 'Bangalore',
  rating: 4.8
}

```

```

{
  _id: ObjectId('6a34182b76d019eb215bcd46'),
  agent_id: 3,
  agent_name: 'Sneha',
  city: 'Chennai',
  rating: 4.9
}

```

```

db.delivery_agents.aggregate([
  {
    $group:{
      _id:null,
      avgRating:{$avg:"$rating"}
    }
  }
])

```

```

{
  _id: null,
  avgRating: 4.766666666666667
}

```

```

db.delivery_agents.aggregate([
  {
    $group:{
      _id:null,
      maxRating:{$max:"$rating"}
    }
  }
])

```

```
])
{
  _id: null,
  maxRating: 4.9
}
db.delivery_agents.aggregate([
  {
    $group:{
      _id:null,
      minRating:{$min:"$rating"}
    }
  }
])
{
  _id: null,
  minRating: 4.6
}
db.delivery_agents.aggregate([
  {
    $group:{
      _id:"$city",
      count:{$sum:1}
    }
  }
])
{
  _id: 'Bangalore',
  count: 1
}
{
  _id: 'Chennai',
  count: 1
}
{
```

```

    _id: 'Hyderabad',
    count: 1
}
db.delivery_agents.aggregate([
  {
    $group:{
      _id:"$city",
      count:{$sum:1}
    }
  },
  {
    $match:{
      count:{$gt:1}
    }
  }
])

```

Reviews

```

use food_delivery_db
switched to db food_delivery_db
db.reviews.insertMany([
  {
    review_id:1,
    customer_id:1,
    restaurant_id:101,
    rating:5,
    comment:"Excellent service"
  },
  {
    review_id:2,
    customer_id:2,
    restaurant_id:102,
    rating:4,
    comment:"Good taste"
  }
])

```

```
},  
{  
  review_id:3,  
  customer_id:3,  
  restaurant_id:103,  
  rating:5,  
  comment:"Very tasty"  
}  
]);
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6a3418543b87a7fe48f47d34'),  
    '1': ObjectId('6a3418543b87a7fe48f47d35'),  
    '2': ObjectId('6a3418543b87a7fe48f47d36')  
  }  
}
```

```
db.reviews.find()
```

```
{  
  _id: ObjectId('6a3418543b87a7fe48f47d34'),  
  review_id: 1,  
  customer_id: 1,  
  restaurant_id: 101,  
  rating: 5,  
  comment: 'Excellent service'  
}
```

```
{  
  _id: ObjectId('6a3418543b87a7fe48f47d35'),  
  review_id: 2,  
  customer_id: 2,  
  restaurant_id: 102,  
  rating: 4,  
  comment: 'Good taste'
```

```

}
{
  _id: ObjectId('6a3418543b87a7fe48f47d36'),
  review_id: 3,
  customer_id: 3,
  restaurant_id: 103,
  rating: 5,
  comment: 'Very tasty'
}
db.reviews.find({
  rating:5
})
{
  _id: ObjectId('6a3418543b87a7fe48f47d34'),
  review_id: 1,
  customer_id: 1,
  restaurant_id: 101,
  rating: 5,
  comment: 'Excellent service'
}
{
  _id: ObjectId('6a3418543b87a7fe48f47d36'),
  review_id: 3,
  customer_id: 3,
  restaurant_id: 103,
  rating: 5,
  comment: 'Very tasty'
}
db.reviews.find({
  rating:{$gt:4}
})
{
  _id: ObjectId('6a3418543b87a7fe48f47d34'),
  review_id: 1,

```



```
customer_id: 1,
restaurant_id: 101,
rating: 5,
comment: 'Excellent service'
}
{
  _id: ObjectId('6a3418543b87a7fe48f47d36'),
  review_id: 3,
  customer_id: 3,
  restaurant_id: 103,
  rating: 5,
  comment: 'Very tasty'
}
db.reviews.find({
  customer_id:1
})
{
  _id: ObjectId('6a3418543b87a7fe48f47d34'),
  review_id: 1,
  customer_id: 1,
  restaurant_id: 101,
  rating: 5,
  comment: 'Excellent service'
}
db.reviews.find().sort({
  rating:1
})
{
  _id: ObjectId('6a3418543b87a7fe48f47d35'),
  review_id: 2,
  customer_id: 2,
  restaurant_id: 102,
  rating: 4,
  comment: 'Good taste'
}
```

```
}
{
  _id: ObjectId('6a3418543b87a7fe48f47d34'),
  review_id: 1,
  customer_id: 1,
  restaurant_id: 101,
  rating: 5,
  comment: 'Excellent service'
}
```

```
{
  _id: ObjectId('6a3418543b87a7fe48f47d36'),
  review_id: 3,
  customer_id: 3,
  restaurant_id: 103,
  rating: 5,
  comment: 'Very tasty'
}
```

```
db.reviews.aggregate([
  {
    $group:{
      _id:null,
      avgRating:{$avg:"$rating"}
    }
  }
])
```

```
{
  _id: null,
  avgRating: 4.666666666666667
}
```

```
db.reviews.aggregate([
  {
    $group:{
      _id:null,
      maxRating:{$max:"$rating"}
    }
  }
])
```

```

    }
  }
])
{
  _id: null,
  maxRating: 5
}
db.reviews.aggregate([
  {
    $group:{
      _id:null,
      minRating:{$min:"$rating"}
    }
  }
])
{
  _id: null,
  minRating: 4
}
db.reviews.aggregate([
  {
    $group:{
      _id:"$restaurant_id",
      reviewCount:{$sum:1}
    }
  }
])
{
  _id: 101,
  reviewCount: 1
}
{
  _id: 102,
  reviewCount: 1
}

```

```

    }
  {
    _id: 103,
    reviewCount: 1
  }
db.reviews.aggregate([
  {
    $group:{
      _id:"$restaurant_id",
      reviewCount:{$sum:1}
    }
  },
  {
    $match:{
      reviewCount:{$gt:1}
    }
  }
])
db.reviews.aggregate([
  {
    $group:{
      _id:"$restaurant_id",
      avgRating:{$avg:"$rating"}
    }
  }
])
{
  _id: 101,
  avgRating: 5
}
{
  _id: 102,
  avgRating: 4
}

```

```

{
  _id: 103,
  avgRating: 5
}
db.reviews.aggregate([
  {
    $group:{
      _id:"$restaurant_id",
      avgRating:{$avg:"$rating"}
    }
  },
  {
    $sort:{
      avgRating:-1
    }
  }
])
{
  _id: 101,
  avgRating: 5
}
{
  _id: 103,
  avgRating: 5
}
{
  _id: 102,
  avgRating: 4
}
db.reviews.aggregate([
  {
    $match:{
      rating:5
    }
  }
])

```

```
    },  
    {  
      $count:"fiveStarReviews"  
    }  
  ])  
  {  
    fiveStarReviews: 2  
  }  
}
```

```
db.reviews.aggregate([  
  {  
    $group:{  
      _id:"$rating",  
      count:{$sum:1}  
    }  
  }  
])  
{  
  fiveStarReviews: 2  
}
```

```
db.reviews.aggregate([  
  {  
    $group:{  
      _id:"$rating",  
      count:{$sum:1}  
    }  
  }  
])  
{  
  _id: 4,  
  count: 1  
}  
{  
  _id: 5,  
  count: 2  
}
```

```
db.reviews.aggregate([  
  {  
    $group:{  
      _id:"$rating",  
      count:{$sum:1}  
    }  
  },  
  {  
    $match:{  
      count:{$gt:1}  
    }  
  }  
])  
{  
  _id: 5,  
  count: 2  
}
```

```
db.reviews.aggregate([  
  {  
    $group:{  
      _id:"$rating",  
      count:{$sum:1}  
    }  
  },  
  {  
    $match:{  
      count:{$gt:1}  
    }  
  }  
])  
{  
  _id: 5,  
  count: 2  
}
```

```

        }
    }
])
{
  _id: 5,
  count: 2
}

```

Advanced Project-Level Aggregation Tasks

```

db.orders.aggregate([
  {
    $group:{
      _id:"$restaurant_id",
      revenue:{$sum:"$total_amount"}
    }
  },
  {
    $sort:{revenue:-1}
  },
  {
    $limit:1
  }
])
{
  _id: 101,
  revenue: 500
}

```

```

db.orders.aggregate([
  {
    $group:{
      _id:"$customer_id",
      orderCount:{$sum:1}
    }
  },
  {

```

```

        $sort:{orderCount:-1}
    },
    {
        $limit:1
    }
])
{
    _id: 3,
    orderCount: 1
}
db.orders.aggregate([
    {
        $group:{
            _id:"$customer_id",
            orderCount:{$sum:1}
        }
    },
    {
        $sort:{orderCount:-1}
    },
    {
        $limit:5
    }
])
{
    _id: 1,
    orderCount: 1
}
{
    _id: 2,
    orderCount: 1
}
{
    _id: 3,

```



```

    orderCount: 1
  }
  db.orders.aggregate([
    {
      $group:{
        _id:null,
        totalOrders:{$sum:1},
        deliveredOrders:
          $sum:{
            $cond:[
              {$eq:["$status","Delivered"]},
              1,
              0
            ]
          }
      }
    },
    {
      $project:{
        percentageDelivered:{
          $multiply:[
            {$divide:["$deliveredOrders","$totalOrders"]},
            100
          ]
        }
      }
    }
  ])
  {
    _id: null,
    percentageDelivered: 66.66666666666666
  }

```

lasAtlas atlas-ku75ct-shard-0 [primary] food_delivery_db

```
db.payments.aggregate([
  {
    $group:{
      _id:"$payment_method",
      count:{$sum:1}
    }
  },
  {
    $sort:{count:-1}
  },
  {
    $limit:1
  }
])
```

```
{
  _id: 'Cash',
  count: 1
}
```

```
db.customers.aggregate([
  {
    $group:{
      _id:"$city",
      count:{$sum:1}
    }
  },
  {
    $sort:{count:-1}
  },
  {
    $limit:1
  }
])
```

```
{
  _id: 'Bangalore',
```

```

    count: 2
  }
  db.restaurants.aggregate([
    {
      $group:{
        _id:"$city",
        count:{$sum:1}
      }
    },
    {
      $sort:{count:-1}
    },
    {
      $limit:1
    }
  ])
  {
    _id: 'Hyderabad',
    count: 2
  }
  db.reviews.aggregate([
    {
      $group:{
        _id:"$restaurant_id",
        avgRating:{$avg:"$rating"}
      }
    },
    {
      $sort:{avgRating:-1}
    },
    {
      $limit:3
    }
  ])

```

```

{
  _id: 103,
  avgRating: 5
}
{
  _id: 101,
  avgRating: 5
}
{
  _id: 102,
  avgRating: 4
}
db.orders.aggregate([
  {
    $lookup:{
      from:"customers",
      localField:"customer_id",
      foreignField:"customer_id",
      as:"customer"
    }
  },
  {
    $unwind:"$customer"
  },
  {
    $group:{
      _id:"$customer.city",
      avgAmount:{$avg:"$total_amount"}
    }
  }
])
{
  _id: 'Bangalore',
  avgAmount: 500
}

```

```
}  
  
{  
  _id: 'Hyderabad',  
  avgAmount: 300  
}  
  
{  
  _id: 'Chennai',  
  avgAmount: 360  
}
```